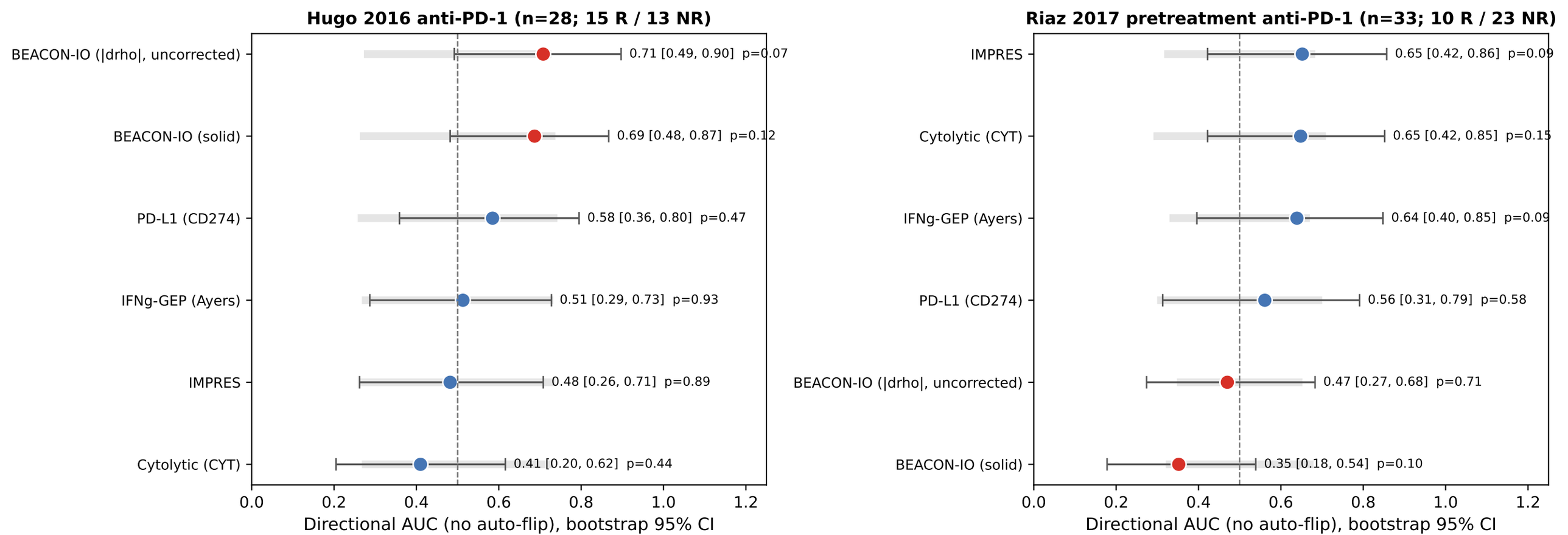

Grey bands: empirical random-gene null (95th percentile of |AUC-0.5|). No signature, established or BEACON-IO (red), exceeds its null in either cohort: these sample sizes cannot adjudicate biomarker performance.